

Frédéric Andrian

Software Engineer · Python, APIs and AI Systems

Bordeaux, France

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Professional Summary

Early-career Software Engineer completing an MSc-equivalent engineering degree in Artificial Intelligence and Cognitive Science. At INRIA, I independently delivered an end-to-end AI application from a user need to a functional solution, including Python and FastAPI services, REST APIs, MongoDB, a React and Next.js interface, Dockerized components, and GitHub CI/CD. I have also operated LLMs on a local Mac Studio, evaluated models on an HPC cluster, and deployed software on a Raspberry Pi. Curious and autonomous, I want to build reliable software for AI workloads across varied infrastructure. Available from November 2026, I am motivated to work abroad to discover new working methods and grow within an international environment.

Experience

INRIA (French National Institute for Research in Digital Science and Technology), Mnémosyne Team

Bordeaux, France

AI Research Engineer Intern

Mar. 2026 – Sep. 2026

- Independently designed and developed, with guidance from my supervisors, an end-to-end multi-step AI application that helps researchers discover scientific papers.
- Built asynchronous Python and FastAPI backend services, REST APIs documented with OpenAPI, and MongoDB persistence.
- Developed a React and Next.js interface that enabled researchers to interact with the application and its tools.
- Built the agent without an orchestration framework to understand and control tool execution, state, memory, context, and multi-step behavior.
- Integrated the OpenAI API and designed the tools used by the agent to retrieve information and complete successive tasks.
- Implemented a RAG pipeline using document chunking, embeddings, semantic search, and ChromaDB.
- Dockerized selected components and implemented a GitHub CI/CD pipeline to automate integration.
- Managed a Dockerized web application for LLMs installed locally on a Mac Studio, including model installation and configuration.
- Developed a Python tool to automatically collect scientific paper metadata from multiple public sources.
- Implemented webhooks to relay the results of scheduled tasks.
- Evaluated OpenCLAW in an isolated Docker environment with attention to the reliability and security of autonomous tool and file access.
- Benchmarked Qwen, Gemma, and GLM on an HPC cluster with Slurm across latency, TTFT, response quality, and context size.
- Evaluated multiple quantization configurations to identify the best trade-off between inference performance and response quality for each execution environment.
- Defined an experimental protocol, analyzed the results, and contributed to a scientific manuscript documenting the model evaluation.

Karunya University

Coimbatore, India

Reinforcement Learning Research Intern

Jun. 2025 – Sep. 2025

- Designed and compared deterministic path planning with a reinforcement learning approach for autonomous navigation of a hospital robot.
- Deployed and optimized the software and reinforcement learning model on an embedded Raspberry Pi.

Technical Skills

Languages and Backend	Python, Java, FastAPI, async/await, Node.js, REST APIs, OpenAPI
Data	MongoDB, SQL, ChromaDB, embeddings, semantic search
Frontend	React, Next.js
Platform and Deployment	Linux, Docker, GitHub CI/CD, Slurm, HPC environments, Raspberry Pi, local LLM deployment
AI Systems	LLMs, OpenAI API, framework-free multi-step agents, tool design, RAG, PyTorch, Deep Learning
Model Evaluation	Qwen, Gemma, GLM, quantization, latency, TTFT, response quality, context size
Development Tools	Git/GitHub, ChatGPT for AI-assisted development
Languages	French (native), English (professional, IELTS 7.5), Spanish (basic)

Education

ENSC, Bordeaux INP

Bordeaux, France

Engineering Degree, MSc-equivalent, in Artificial Intelligence and Cognitive Science

2023 – 2026

- Coursework in software engineering, machine learning, large language models, natural language processing, deep learning, and human-computer interaction.